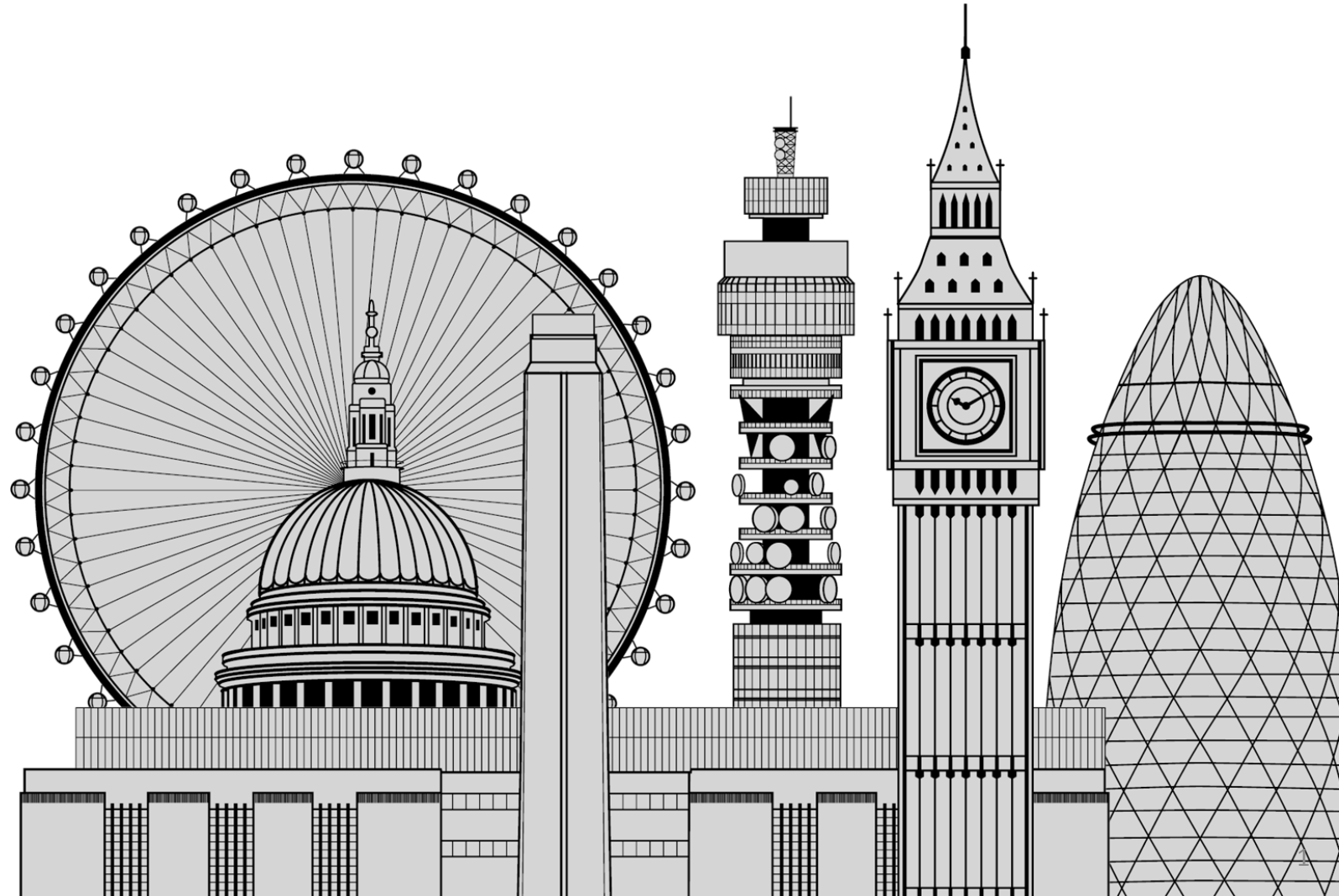


5COSC024W Server-Side Web Development

Part 01: Using PHP & MySQL for Back-end Web Development

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Part 01: Outline

○ **Back-end Web Development**

- Aim, purpose & focus.
- Roles of a back-end web developer.
- Tech overview.

○ **Static web pages**

- HTML.
- CSS.

○ **Dynamic web pages**

- Client-side scripting & JavaScript.
- Server-side scripting & PHP.

○ **Web-Database integration with relational databases**

- Using server-side scripting.
- Requirements.

Back-end Web Development: aim, purpose, focus & roles

— ○ **Aim:**

- It lays the foundational code that enables websites to process the actions that users take on the front-end and deliver the correct information in return.

○ **Purpose:**

- Handles the 'behind-the-scenes' functionality of web apps.
- Manages user connections.
- Powers the web application (i.e. interface).
- Connects the web to a databases.

○ **Focus:**

- How web app works.
- Functionality & logic powering the application i.e. tech never seen by the user.

○ **Roles of a back-end web developer:**

- Writing APIs & libraries.
- Working on business processes and data architecture.
- Writing code to interact with a database.

Back-end Web Development: an overview of the tech



Static web page



- (1) Administrator writes HTML and saves on server.**
- (2) User makes request for static page.**
- (3) Web server locates HTML file.**
- (4) Web server sends stream of HTML to the client.**
- (5) Web browser interprets HTML and displays static page.**

Static web pages

- Delivered to the user's web browser exactly as stored.
- Displays the same info for all users independently from when it is accessed.
- Written in HTML stored as files and made available by the web server over HTTP.
- Offers reasonable performance and security at a low cost, particularly on cloud.
- Limited in terms of functionality and user's interactivity.

HTML (Hyper Text Markup Language)

- Standard markup language for creating web pages.
- Describes the structure of a web page.
- Consists of a series of elements that tell a browser how to display the content which are used as building blocks of the page.
- HTML elements label pieces of content such as "heading", "paragraph", "breaking line", "table", "link", "form element", etc.
- A web browser receives an HTML document from a web server and render the document into multimedia pages.

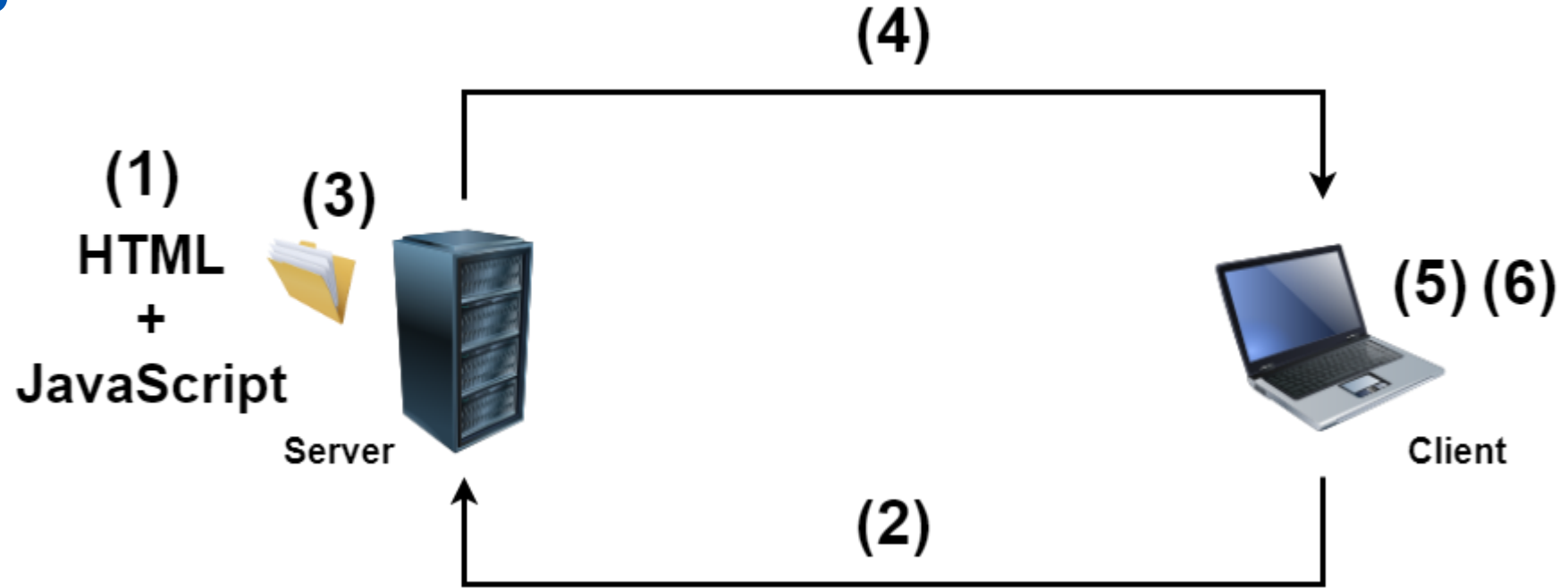
CSS (Cascading Style Sheets)

- Standard style sheet language used for describing the presentation of a document written in a markup language such as HTML by formatting elements such as layout, colours, and fonts.
- Describes how HTML elements are to be displayed on screen.
- Enables the separation of presentation and content.
- Allows the centralisation of presentation elements in separate file, the sharing of formatting by multiple pages and edits to be made for the entire web app.
- Can control the layout of multiple web pages all at once.
- Ensures greater consistency in the look-and-feel of the app.
- External stylesheets are stored in CSS files.

Dynamic web pages

- Content is generated each time it is accessed.
- Can be customised by and for each user.
- Can respond to user's input from browser.
- Requires different technologies that can perform conversions from different data formats into HTML 'on the fly' either on the server or on the browser.
- Can interconnect with a back-end database.

Dynamic web page: client-side

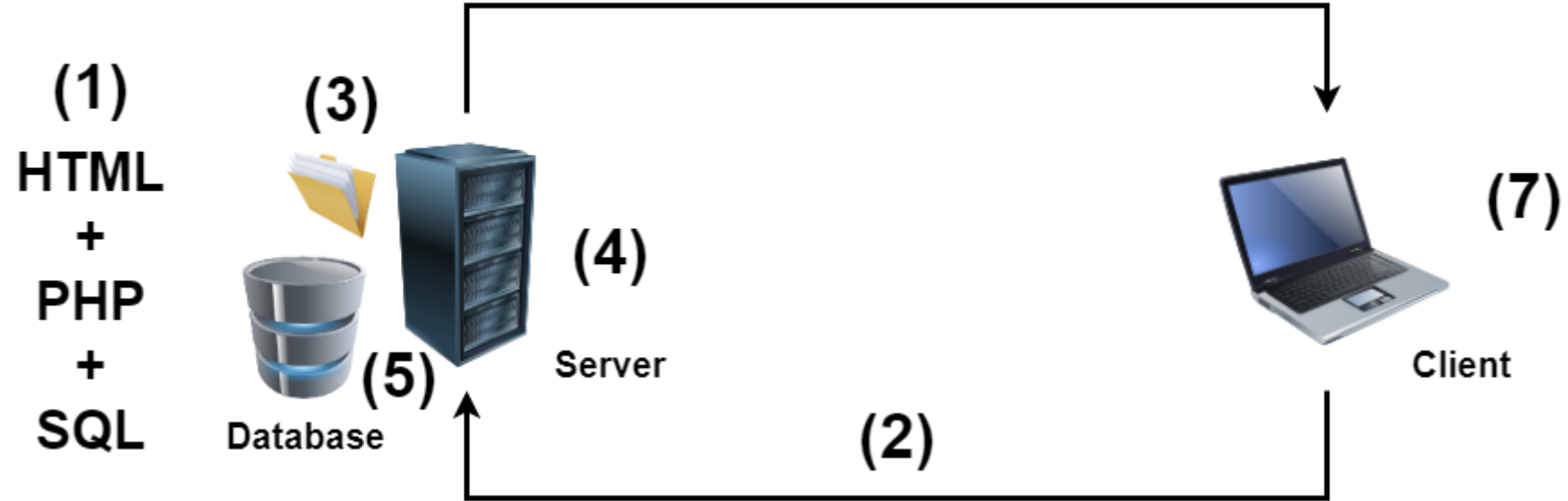


- (1) Administrator writes HTML with JavaScript and saves on server.
- (2) User makes request for dynamic page.
- (3) Web server locates HTML file containing the JavaScript.
- (4) Web server sends HTML and JavaScript to the client.
- (5) Web browser processes the JavaScript and generates HTML.
- (6) Web browser interprets HTML and displays web page.

JavaScript: initially just a client-side scripting language

- Initially a browser client-scripting language
- Browser as host environment allows:
 - Capture of user's input.
 - Making of HTTP requests.
 - Processing of JavaScript instructions to output HTML and CSS.

Dynamic web page: server-side



- (1) Administrator writes HTML with embedded PHP and SQL & saves on server.
- (2) User makes request for dynamic page.
- (3) Web server locates HTML file containing the PHP script and the SQL query.
- (4) Web server gets the script engine to process the PHP and generate HTML.
- (5) Script engine connects with the database and runs SQL query.
- (6) Web server sends stream of HTML to the client.
- (7) Web browser interprets HTML and displays web page.

Server-side Scripting Languages

- Server-Side Scripting languages used to extend browser and Web server with database functionality.
- Script code embedded in HTML & downloaded every time page accessed.
- Updating browser is simply a matter of changing file on server.
- Most popular scripting languages are: PHP, ASP, Java, Ruby, Perl.
- They are all interpreted languages, not compiled, making it easy to create small applications.

PHP (Hypertext Preprocessor)

- PHP: open source scripting language developed in the mid 1990's.
- Intended to be embedded within HTML pages.
- When PHP page is served, HTML portions are sent directly back to the Web browser.
- Any code contained within special PHP tags `<?php` and `?>` are executed by the PHP parser
- Results of the execution (stream of HTML) returned to the Web browser.

Web-DB Integration: server-side scripting & relational DB

○ Web Database Integration (relational DB)

- Web Server interacts with DBMS using SQL.
- In response, DBMS returns a set of records or an error message.
- Web Server generates appropriate Web page and sends it back to Web Browser.

○ Example

- Example: link to display a list of all films in particular genre.
https://www.coolfilmsite.com/film_list.php?genre=drama
- Web server calls a server-side script, written in PHP and pass along the parameter “genre=drama”.
- At Web server, request unpacked, formed into a SQL query, and submitted to DBMS on database server.
- In response, DBMS returns set of records to Web server, which then apply HTML tags to the records so that they display properly in Web browser.

Requirements for Web-DBMS Integration

- Support for transactions that span multiple HTTP requests.
- Support for session-based and application-based authentication.
- Acceptable performance.
- Minimal administration overhead.
- Set of high-level productivity tools to allow applications to be developed, maintained, and deployed with relative ease and speed.